Saugandhi Nyshadham

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Work authorization: German student visa — 20h/week during semester; full-time during semester breaks. Open to Nuremberg, Erlangen, Fürth, Munich, and remote/hybrid.

PROFILE

M.Sc. Artificial Intelligence student at FAU with an applied statistics and data analytics background. Builds end-to-end data and ML systems — from ingestion and curation to model evaluation and interpretability. Shipped an offline-first e-ink study dashboard on Raspberry Pi. Seeking Werkstudent, internship, or working-student roles in data science, machine learning, or AI engineering. Available 20h/week during semester.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) **03/2026 – Present**
M.Sc. Artificial Intelligence — Minor: AI in Biomedical Engineering *Expected 03/2028*

- Core: Subsymbolic AI, Symbolic AI, AI Systems.
- **P**AIR seminar

OTH Amberg-Weiden **03/2025 – 02/2026**
M.Sc. AI for Industrial Applications

- Coursework: Industry 4.0, Medical Systems Engineering.

Amrita Vishwa Vidyapeetham **09/2022 – 07/2024**
M.Sc. Applied Statistics & Data Analytics

- Coursework: Machine Learning, Deep Learning, Statistical Modeling, Data Analytics.
- Thesis: Explainable DiscNN for DoH Analysis and Network Characterization.

WORK EXPERIENCE

Junior Data Analyst — SBV Solutions **07/2024 – 03/2025**
Hyderabad, India

- Analyzed and curated large datasets powering an internal AI platform, supporting features including Active Learning, Bayesian Statistics, Exploratory Data Analysis, Image Segmentation, and Feature Extraction.
- Built and maintained data pipelines on Google Cloud Platform and Apache Airflow, reducing manual processing steps across recurring workflows.
- Performed exploratory data analysis in Python (Pandas, NumPy) to surface trends and data quality issues, documenting findings for engineering and product stakeholders.
- Collaborated with the platform team to integrate data-driven features into production workflows, improving data reliability and downstream model performance.

Machine Learning Intern — IDeepeners **10/2023 – 11/2023**

- Developed ML/DL models for financial market analysis, improving prediction accuracy on historical data.
- Processed and cleaned large financial datasets to uncover patterns and inform trading strategy optimisation.
- Applied deep learning algorithms to real-world financial data, collaborating with a team of experts on complex analytics problems.

PROJECTS

E-Ink Personal Study Dashboard (Raspberry Pi Zero 2 WH) — *Solo project* **08/2026 – 10/2026**

- Built an offline-first e-ink dashboard for Notes, study Docs, and Calendar on a Raspberry Pi Zero 2 WH with a 7" e-ink HAT, used daily for study and scheduling.
- Designed a phone-to-device sync pipeline so notes written on mobile appear on the e-ink display.

- Implemented PIN lock and an SOS screen exposing emergency health info and lost-and-found details for daily carry.
- Integrated a local, privacy-preserving summarization/tagging model (no cloud, no telemetry), keeping all personal data on-device.

Explainable DiscNN for Traffic Analysis & Interpretability — *Master's thesis*

- Designed a PyTorch CNN to characterize DNS-over-HTTPS traffic as part of a research thesis on interpretable network modeling.
- Applied XAI methods to probe hidden layers and visualize feature importance, identifying features that drove model decisions and improving auditability of the classifier.
- Audited the model against NIST AI RMF MEASURE guidance and conducted a CSET-aligned harm analysis, documenting misclassification risks, severity levels, and affected entities.

BODY-SYNTH: Generative Modeling & Dataset Auditing — *Team project, dataset & model lead*

- Audited and cleaned 5 medical image repositories (WoundsDB, ISIC Archive), reducing distributional anomalies across the dataset.
- Built a Python/OpenCV curation pipeline standardizing images to 512×512 with uniform intensity normalization, cutting preprocessing time.
- Evaluated GANs (BodyGAN) and diffusion models (Stable Diffusion, MedFusion) using CLIP Score and FID, selecting the best-performing model for downstream synthesis.
- Containerized the pipeline with Docker and tracked experiments in MLflow with GitHub Actions CI, making results reproducible across environments.

TECHNICAL SKILLS

- **Languages:** Python, R, SQL
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, OpenCV, GANs, Diffusion Models, XAI, Model Probing
- **Data/Signal:** Pandas, NumPy, SciPy, Matplotlib, Seaborn
- **ML Platform:** Google Cloud Platform (VMs), Apache Airflow
- **MLOps:** Docker, MLflow, GitHub Actions
- **Tools:** Git, GitHub, MySQL

LEADERSHIP & EXTRACURRICULAR

TUM Opportunity Hunters **2026 – Present**
Participant

- Selected for a TUM program focused on identifying and pursuing high-impact career and research opportunities.

AI Safety **2026 – Present**
Member

- Lens Academy AI Risk Fundamentals
- Member of AI Safety Nuremberg

Medium **2025 – Present**
Writer

- Published 2 mini-articles on AI, politics, and society, communicating technical topics to a general audience.

LANGUAGES

English C1

German A2 (actively learning, B1 exam target: Dec 2026)

Telugu, Hindi native